



SCMS School of Engineering and Technology
Department of Computer Science and Engineering
Seminar Proposal Form
CSQ 413 SEMINAR BATCH 2023-2027

Name & Reg No. of Student: Pranav Jayan, SCM23CS209

Class: S7 B.Tech CS4

Date: 13-08-2026

1. Area of seminar topic:

Large Language Model (LLM) Agents and Tool-Augmented Reasoning for Efficient Task Solving

2. Title of Seminar:

A Joint Optimization Framework for Enhancing Efficiency of Tool Utilization in LLM Agents

3. Literature Referred:

Sl No	Journal Name	Paper Title	Authors	Month & Year	ISSN/Issue/Volume	Page No
1	Findings of ACL (ACL Anthology)	A Joint Optimization Framework for Enhancing Efficiency of Tool Utilization in LLM Agents	B. Wu, E. Meij, E. Yilmaz	July 2025	pp. 22361–22373	22361-22373
2	ICLR (Conference)	ReAct: Synergizing Reasoning and Acting in Language Models	S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, Y. Cao	May 2023	arXiv:2210.03629	—
3	NeurIPS (Conference)	Toolformer: Language Models Can Teach Themselves to Use Tools	T. Schick, J. Dwivedi-Yu, R. Dessì	December 2023	NeurIPS 36	—
4	NeurIPS (Conference)	Gorilla: Large Language Model Connected with Massive APIs	S. G. Patil, T. Zhang, X. Wang, J. E. Gonzalez	December 2024	NeurIPS 38	—
5	ICLR (Conference)	ToolLLM: Facilitating Large Language Models to Master 16000+ Real-world APIs	Y. Qin, S. Liang, Y. Ye, K. Zhu, L. Yan, Y. Lu	May 2024	ICLR 2024	—

4. Seminar Abstract:

As Large Language Model (LLM) agents are increasingly deployed to solve complex, multi-step tasks by invoking external tools and APIs, the efficiency of tool utilization has become a critical bottleneck. Existing approaches typically rely on an LLM agent that follows instructions in the agent prompt and consults tool descriptions to decide which tool to call, but this decision-making process is often supplemented with inference-scaling techniques such as chain-of-thought and tree-of-thought reasoning. While effective, these techniques introduce substantial computational overhead, making them impractical for real-world, latency-sensitive applications.

This seminar presents a joint optimization framework that addresses this efficiency gap by formalizing the combined role of the instructions provided in the agent prompt and the tool descriptions available to the agent — collectively referred to as "context." The framework identifies incomplete or poorly aligned context as a key driver of unnecessary computational overhead in tool-augmented LLM agents, and proposes an optimization procedure that jointly refines both the agent-side instructions and the tool-side descriptions so that they interact more effectively, rather than optimizing each in isolation.

The approach is evaluated on StableToolBench and RestBench, two established benchmarks for tool-augmented language agents, and is shown to achieve superior efficiency — in terms of reduced reasoning steps and computational cost — while maintaining, and in several cases improving, task-solving effectiveness compared to baseline prompting strategies. The seminar will situate this work within the broader landscape of tool-use methods for LLM agents, including reasoning-and-acting paradigms such as ReAct, self-supervised tool-use learning as in Toolformer, and large-scale API-grounded agents such as Gorilla and ToolLLM, to highlight how context optimization complements and improves upon these prior directions.

This seminar will discuss the formalization of agent context, the joint optimization methodology for refining prompts and tool descriptions, the experimental setup and benchmarking methodology, comparative results against existing baselines, and the practical implications of this workflow.

Approved by

Seminar Guide	Seminar Coordinator	HOD
Ms. Surya SG	Ms. Surya SG	Dr. Manish TI
Asst. Professor	Asst. Professor	Professor
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